

$Q^T = Q^{-1} \Leftrightarrow QQ^T = Q^T Q = I \Leftrightarrow$ cols orthonormal
 unitary preserve: $(Qx)^*(Qy) = x^*y$
 $\|Qx\| = \|x\|$
 Cauchy Sw: $\langle x, y \rangle \leq \|x\| \|y\|$
 $\|A\|_{\infty, n} := \sup_{\|x\|_\infty=1} \|Ax\|_\infty$
 $\|AB\|_{\infty, n} \leq \|A\|_{\infty, m} \|B\|_{\infty, n}$
 $\|A\|_1 = \max_{1 \leq j \leq n} \|a_j\|_1$ (=max abs col sum), $\|A\|_2 = \sigma_1$
 $\|A\|_\infty = \max$ abs row sum
 $\|A\|_F = \sqrt{\sum \sum |a_{ij}|^2} = \sqrt{\sum \|a_j\|_2^2} = \sqrt{\text{tr}(A^*A)} = \sqrt{\sum \sigma_i^2}$
 $\|AB\|_F^2 \leq \|A\|_F^2 \|B\|_F^2$, $\|QA\|_2 = \|A\|_2$, $\|QA\|_F = \|A\|_F$

projector $P: P^2=P \Leftrightarrow P^*=P$ (not ortho) (ortho)
 P projector $\Rightarrow I-P$ also projector
 $V \xrightarrow{\text{along } S_1} S_2 \quad V \xrightarrow{\text{along } S_2} S_1$
 P orthogonal projector $\Leftrightarrow S_1 \perp S_2 \Leftrightarrow P=P^*$
 P orthogonal projector $\Rightarrow I-P$ also orthogonal projector
 orthogonal proj $P \Rightarrow P=Q\Sigma Q^*$ both SVD and eigendecomp
 $\{q_1, \dots, q_n\}$ span S_1 , $\{q_{n+1}, \dots, q_m\}$ span S_2
 $\Sigma: \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ & & 0 \end{bmatrix}$
 can just take $[q_1, \dots, q_n]$ as $\hat{Q} \Rightarrow P = \hat{Q}\hat{Q}^*$ (orthogonal proj onto $\text{im}(\hat{Q})$)
 specially, proj onto line of $\langle a \rangle$: $P_{aa} = \frac{aa^*}{a^*a}$
 Given arbitrary basis $\{a_1, \dots, a_n\}$, $P_{\langle a_1, \dots, a_n \rangle} = A(A^*A)^{-1}A^*$

SVD: $V_i \mapsto \sigma_i U_i$
 reduced: $A = \hat{U} \hat{\Sigma} \hat{V}^*$ full: $A = U \Sigma V^*$
 $\begin{matrix} m \times m & n \times n & m \times n \\ \text{ortho} & \text{diag} & \text{ortho} \end{matrix}$
 $\begin{matrix} m \times m & m \times n & n \times n \\ \text{uni} & \text{diag} & \text{uni} \end{matrix}$
 solve $Ax=b$ by SVD: $A=U\Sigma V^*$, $x := V^*x$, $b := U^*b \Rightarrow b' = \Sigma x'$
 1. $\text{rank}(A) = \langle u_1, \dots, u_r \rangle$, $\text{null}(A) = \langle v_{r+1}, \dots, v_n \rangle$
 2. nonzero $\sigma(A)$ is $\sqrt{\lambda(A^*A)}$
 3. $\det A = \prod \sigma_i$
 4. $A = \sum_{j=1}^r \sigma_j U_j V_j^*$
 $\|A - A_{UV}\|_2 = \sigma_{r+1}$
 $\|A - A_{UV}\|_F = \sqrt{\sigma_{r+1}^2 + \dots + \sigma_n^2}$

QR $A = QR$ $\begin{matrix} m \times m & m \times n & m \times n & n \times n \\ Q & R & \vec{a}_n = \sum_{j=1}^n r_{jn} \vec{e}_n \end{matrix}$
 QR def

Classical GS: $v_j = \sum_{i=1}^j q_i q_i^* q_j$
 for $j=1$ to n
 $v_j = q_j$
 for $i=1$ to $j-1$
 $r_{ij} = q_i^* q_j$
 $v_j = v_j - r_{ij} q_i$
 $r_{jj} = \|v_j\|_2$
 $q_j = v_j / r_{jj}$ (normalize)
 $v_j = \text{proj}_{\langle v_1, \dots, v_{j-1} \rangle} v_j$
 $P = I \Rightarrow P_j$
 ex. (classical GS)
 $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
 $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
 $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
 $q_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
 $q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
 $\hat{Q} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$
 $\hat{R} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$

Householder
 for $k=1$ to n
 $x = A_{k:m, k}$
 $v_k = \text{sign}(x_1) \|x\|_2 e_1 + x$
 $v_k = \frac{v_k}{\|v_k\|_2}$
 $A_{k:m, k:n} = A_{k:m, k:n} - 2v_k v_k^* A_{k:m, k:n}$
 eigenvalue decomp: $A \in \mathbb{C}^{n \times n}$, $A = X \Lambda X^{-1}$, X eigenvectors
 A diagonalizable ($GM = AM$)
 $(A - \lambda I)x = 0$

implicit calculate Q^*
 for $k=1$ to n
 $b_{k:m} = b_{k:m} - 2v_k v_k^* b_{k:m}$
 implicit calculate Qx
 for $k=n$ to 1
 $x_{k:m} = x_{k:m} - 2v_k v_k^* x_{k:m}$

Modified GS:
 for $i=1$ to n
 $v_i = a_i$
 for $i=1$ to n
 $r_{ii} = \|v_i\|$
 $q_i = v_i / r_{ii}$
 for $j=i+1$ to n
 $r_{ij} = q_i^* v_j$
 $v_j = v_j - r_{ij} q_i$
 work: $\sim 2mn^2$ flops
 matrix form:
 $A R_1 R_2 \dots R_n = \hat{Q}$
 $\hat{R}^{-1}$
 $R_1 = \begin{bmatrix} 1 & r_{12} & r_{13} & \dots \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$
 $R_2 = \begin{bmatrix} 1 & & & \\ & 1 & r_{23} & \dots \\ & & \ddots & \\ & & & 1 \end{bmatrix}$

$Q_k = \begin{bmatrix} I_{k-1} & \\ & F \end{bmatrix}$
 $F = I - 2 \frac{vv^*}{\|v\|^2}$
 eg Q_k reflects $\text{span}(x)$ to $\text{span}(e_1)$
 $\text{span}(x) \rightarrow \text{span}(e_1)$
 $\text{span}(e_1) \rightarrow \text{span}(e_1)$
 $Q_1 \dots Q_n Q_n^* A = R$
 $= Q^*$

To get full QR: find $b_3 \perp \text{im}(A)$, normalize it
 $Q = [\hat{Q} \ b_3]$
 $\hat{R} = [R \ 0]$

Least squares
 $\min \|r\|_2 \Leftrightarrow A^* r = 0 \Leftrightarrow A^* A x = A^* b \Leftrightarrow \text{proj}_A b = Ax$
 use SVD to solve least sq.
 $A = \hat{U} \hat{\Sigma} \hat{V}^*$ note: $\text{im}(A) = \text{im}(\hat{U})$, $A^* = \hat{U}^* \hat{\Sigma}^* \hat{V}^*$
 $\Rightarrow Ax_{\text{op}} = \hat{U} \hat{\Sigma}^* \hat{U}^* b$ (or $\hat{U} \hat{\Sigma}^*$)
 $x_{\text{op}} = A^+ b = \hat{V} \hat{\Sigma}^+ \hat{U}^* b$
 $\hat{\Sigma}^+$ pseudoinverse
 use QR of A to solve least sq.
 $\text{proj}_A b = \hat{Q} \hat{Q}^* b = Ax \Rightarrow \hat{R} x = \hat{Q}^* b \Rightarrow x = \hat{R}^+ \hat{Q}^* b$

$F_{2n} = \begin{bmatrix} F_n & T_n F_n \\ F_n & -T_n F_n \end{bmatrix}$ $w_n = e^{\frac{2\pi i}{2n}}$
 $T_n = \text{diag}(1, w_n, w_n^2, \dots, w_n^{n-1})$
 F_{2n} is 重排 F_n 的列, by 逆序排列
 ex $\begin{matrix} 0 = 0_2 & 1 = 0_2 & 2 = 1_2 & 3 = 1_2 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 0 = 0_2 & 2 = 1_2 & 1 = 0_2 & 3 = 1_2 \end{matrix}$
 ex $F_4 = \begin{bmatrix} F_2 & [-1] F_2 \\ F_2 & [-1] F_2 \end{bmatrix}$
 Sineigenvalue: $\langle \lambda \rangle_{k=0}^{n-1}$, $\lambda_k = e^{\frac{2\pi i k}{n}}$
 $\Rightarrow F_n^* S_n F_n = D_n = \begin{bmatrix} w^0 & & \\ & \ddots & \\ & & w^{n-1} \end{bmatrix}$
 $F_n^* (P(S_n)) F_n = (P(D))$
 for $p(z) = \sum_{j=0}^{n-1} c_j z^j = c_0 + c_1 z + c_2 z^2 + \dots + c_{n-1} z^{n-1}$
 Define: $P_{\text{even}}(z) = c_0 + c_2 z^2 + \dots + c_{n-2} z^{\frac{n-2}{2}}$
 $P_{\text{odd}}(z) = c_1 + c_3 z + \dots + c_{n-1} z^{\frac{n-1}{2}}$
 $\Rightarrow p(z) = P_{\text{even}}(z^2) + z P_{\text{odd}}(z^2)$
 $= \frac{(1+z^2)}{2} P_{\text{even}}(z) + \frac{(1-z^2)}{2} P_{\text{odd}}(z/w_n)$
 ex $n=4$, $p(z) = a + bz + cz^2 + dz^3$
 Given $p(1)=2$, $p(i)=3$, $p(-1)=5$, $p(-i)=8$
 $\hat{a}, \hat{b}, \hat{c}, \hat{d}$
 Sol $p(z) = \frac{1+z^2}{2} P_{\text{even}}(z) + \frac{(1-z^2)}{2} P_{\text{odd}}(z/w_n)$
 note: $P_{\text{even}}(z)$ is (1,2) and (1,4) $\Rightarrow P_{\text{even}}(z) = \frac{1+z^2}{2}$
 $P_{\text{odd}}(z) = \frac{1+z^2}{2} - 3 + \frac{1+z^2}{2} = \frac{1+z^2}{2} - 5$
 $\Rightarrow p(z) = \frac{1+z^2}{2} - 5 + \frac{(1-z^2)}{2} (11-5i)$